Do Cointegration-based Pairs Remain Profitable? *

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Abstract

This study examines the out-of-sample profitability of cointegration-based formation across choice dimensions including combination methods, stock groupings, and window lengths, while testing whether formation-stage mean reversion persists. Pairs are formed from S&P 500 constituents by 5 cointegration-based methods within sectors and correlation-based clusters, and are traded by a 2-standard-deviation threshold rule. The results show that no formation choice can consistently generate profit on its own. The pairs with the strongest formation evidence lose between 0.7% and 1.3% annually in the testing sample. Their mean reversion persists little more than that of spreads formed on independent random walks. The result agrees with the decline in pairs trading profits reported since 2010.

1 Introduction

A pair trade exploits long-run economic equilibria by taking offsetting positions in assets when temporary price deviations occur. The strategy involves two phases, a formation step that selects assets and specifies spreads, and a trading step that opens positions upon divergence and closes them upon mean reversion. Cointegration serves as the standard formation tool, identifying stationary linear combinations of nonstationary prices [1, 2]. This framework extends to baskets through Johansen cointegration [3] and the principal components (PCs) [4], as well as across asset groups defined by clustering algorithms [5].

However, prior empirical literature documents declining profitability. Clegg and Krauss [6] report a net return of -2.76% a year from 2010 to 2015, and Bui and Ślepaczuk [7] find only 0.06% a year for Engle–Granger pairs on the Nasdaq 100 from 2000 to 2021. Reichmann [8] reports a cumulative net loss of 5.49% for pairs selected by price distance from 2020 to 2023. Cointegration found at formation also rarely persists. The share of pairs that remain cointegrated a year later is about the size of the test [9], and Farago and Hjalmarsson [10] show that truly cointegrated stocks would imply Sharpe ratios above 10, far beyond those observed.

This study evaluates the out-of-sample profitability of cointegration-based formation across choice dimensions, including combination methods, stock groupings, and window lengths, and tests whether formation-stage mean reversion persists. Using S&P 500 daily data from 2010 to 2025, we examine 5 combination methods across sectors and correlation-based clusters over formation windows of 1 to 5 years. The threshold rule of Gatev et al. [11] is applied over horizons of 1 to 12 months against an independent random walk null.

We find that no formation choice can consistently generate profit on its own. In the testing period from 2020 to 2025, the top decile of every method loses between 0.70% and 1.34% annually. The formation p-value does not consistently rank returns, and neither grouping nor window changes the sign of the result. The mean reversion detected at formation persists little more than that of spreads formed on random walks.

The remainder of the report is organized as follows. Section 2 describes the data. Section 3 states the formation, the random walk null and the trading rule. Section 4 reports the results, and Section 5 concludes.

*Code and replication data are available at *GitHub*.

2 Data

Since this study analyzes data from 2010 to 2025, when the S&P 500 evolved through mergers, spin-offs, and market-cap adjustments, the dataset includes the S&P 500's point-in-time membership [12]. Daily adjusted closing prices from 2005 to 2025 are fetched from Yahoo Finance, and a stock is eligible if it has a complete price record over the formation window. Formation dates are the first trading days of each month. The training sample comprises the trading windows that start between 2010 and 2019, and the testing sample those that start between 2020 and 2025.

3 Methodology

3.1 Spreads and Grouping

Write $P_{i,t}$ for the adjusted closing price of stock i on day t. The formation window Φ covers the $L \in \{1, \ldots, 5\}$ years before a formation date. The trading window E starts on the formation date and lasts $H \in \{1, 3, 6, 12\}$ months. With normalized prices $\tilde{p}_{i,t} = P_{i,t}/P_{i,t_0}$, where t_0 is the last day of Φ, we define a spread of m stocks as

$$s_t = \sum_{i=1}^{m} w_i \, \tilde{p}_{i,t}, \qquad \|w\|_1 = 1, \tag{1}$$

with weights estimated on Φ and held fixed over E. Here, we compare three groupings.

Sectors. Each of the 11 sectors is one group.

OPTICS clusters. The market mode is first removed from daily returns by a regression on the equal-weighted return of all eligible stocks [13]. Within each sector, we represent each stock by its PC loadings above the noise bound of random matrix theory [14], then cluster the stocks with OPTICS [15] following [5]. OPTICS does not require specifying the number of clusters and leaves stocks that fit no cluster unassigned. In the training sample, this gives about 24 clusters and leaves 43% of the stocks unassigned and excluded.

Complete-linkage clusters. Within each sector, the returns are clustered by complete linkage on a correlation distance. The number of clusters equals the number of PCs above the noise bound. In the training sample, this gives 31 to 45 clusters on average, depending on L.

3.2 Combination Methods

Method	Weights	Formation statistic
Pairs	Regression of one normalized price on the other	Smaller ADF τ
Johansen	First cointegrating vector [3]	Trace statistic
PCA	Smallest PC of normalized price levels [4]	ADF τ of the spread
Eigenanalysis	Smallest eigenvector of squared autocovariances sum [16]	ADF τ of the spread
Stock versus group	Regression on the equal-weighted rest of its group	ADF τ of the residual

Table 1: Combination methods using the augmented Dickey–Fuller (ADF) statistic τ. All methods except pairs require a group of $m \geq 3$ stocks. Baskets also require $T/m \geq 6$ [17].

Table 1 lists the 5 combination methods, with the equations in Appendix A.

3.3 Random Walk Null

Standard Dickey–Fuller critical values cannot be applied to the formation statistics, because each method chooses its weights to make the spread look stationary. In two-asset pairs, this reflects the classic residual-based testing problem, where critical values must be simulated on independent random walks to correct for selection bias [1, 18]. This distortion becomes substantially more pronounced in larger baskets. PCs of the levels of independent random walks yield residuals that a Dickey–Fuller test finds stationary in almost every sample [19], and the Johansen test over-rejects when the system is large relative to the sample [17].

Therefore, we adopt the standard simulation framework and apply each method to independent random walks. To construct the null distribution, we model log prices as independent driftless Brownian motions,

$$dX_{i,t} = \sigma_i \, dW_{i,t}, \qquad i = 1, \ldots, m, \tag{2}$$

where $W_{i,t}$ are independent standard Brownian motions, and each σ_i is drawn from the daily return standard deviations from 2005 to 2019 of all stocks with usable prices. Daily paths of length $T = 252L$ are normalized as $\tilde{p}_{i,t} = e^{X_{i,t} - X_{i,t_0}}$. For each method, we simulate 1,000 draws (or 10,000 for pairs).

The p-value p of a spread is the share of null statistics at least as extreme as its own. Spreads are assigned to deciles $d = \min\{\lfloor 10p \rfloor, 9\}$, where $d = 0$ ($d = 9$) denotes the strongest (weakest) formation evidence. The proportion of true nulls among N spreads is estimated as $\hat{\pi}_0 = \min\{1, 2\,\#\{p \geq 0.5\}/N\}$ [20]. Extending simulated paths through the trading window yields empirical null values for all subsequent performance metrics.

3.4 Trading Rule and Measures

Write $z_t = (s_t - \bar{s}_\Phi)/\hat{\sigma}_\Phi$, where $\bar{s}_\Phi$ and $\hat{\sigma}_\Phi$ are the sample mean and standard deviation of the spread over Φ. The position $n_t \in \{-1, 0, 1\}$ set at the close of day t follows the rule in [11],

$$n_t = \begin{cases} -\operatorname{sgn}(z_t) & \text{if } n_{t-1} = 0 \text{ and } |z_t| \geq 2, \text{ and } t \text{ is not the last day of } E, \\ 0 & \text{if } n_{t-1} \neq 0 \text{ and } z_t \, n_{t-1} \geq 0, \text{ or } t \text{ is the last day of } E, \\ n_{t-1} & \text{otherwise.} \end{cases} \tag{3}$$

A position opens when the spread is 2 standard deviations from its mean, and closes when the spread returns to the mean or the window ends. The net return r of a spread is its gain over E on one unit of capital after transaction costs. Trading costs 10 basis points (bps) per unit traded, and borrowing costs 25 bps a year on the legs held short [21].

Persistence is measured in two ways. The first is the ADF statistic for the spread over the formation and trading windows, written τ_Φ and τ_E. The second is the re-test, which passes when τ_E lies below the 5% critical value in [18].

3.5 Portfolios and Inference

For each method, write $\bar{r}_{g,L,k}$ for the mean net return of the top-decile spreads of grouping g and window L in trading period k. With $\mathcal{C}_k$ the set of (g, L) that hold at least one top-decile spread in period k, the method portfolio return is the equal-weighted average

$$r_k = \frac{1}{|\mathcal{C}_k|} \sum_{(g,L) \in \mathcal{C}_k} \bar{r}_{g,L,k}. \tag{4}$$

4 Results

4.1 Formation Evidence and Persistence

Table 2 shows that formation evidence for pairs and stock-versus-group spreads is virtually indistinguishable from pure chance, with estimated true-null shares ($\hat{\pi}_0$) ranging from 0.95 to 1.00. In contrast, basket methods yield lower $\hat{\pi}_0$ estimates (0.66 to 0.77), indicating formation statistics that exceed null expectations. Even under the null, basket statistics lie far from the Dickey–Fuller values, with τ_Φ between -7.2 and -8.1, because optimizing across multiple independent series spuriously induces stationarity by construction [19].

			τ_Φ		τ_E		Re-test pass (%)	
Method	$\hat{\pi}_0$	Spreads	Data	Null	Data	Null	Data	Null
Training, 2010 to 2019								
Pairs	1.00	6014	−3.77	−3.78	−1.57	−1.51	5.7	4.8
Johansen	0.70	84	−7.70	−7.17	−1.61	−1.51	5.9	4.8
PCA	0.66	84	−7.78	−7.64	−1.66	−1.53	7.2	5.2
Eigenanalysis	0.68	83	−7.90	−7.75	−1.65	−1.50	6.6	4.7
Stock versus group	0.95	592	−3.58	−3.57	−1.54	−1.51	5.0	5.1
Testing, 2020 to 2025								
Pairs	1.00	7579	−3.75	−3.78	−1.63	−1.51	5.5	4.8
Johansen	0.68	116	−8.25	−7.63	−1.69	−1.51	7.2	4.8
PCA	0.76	80	−7.80	−7.65	−1.75	−1.52	8.9	5.1
Eigenanalysis	0.77	84	−8.23	−8.07	−1.75	−1.49	9.0	4.7
Stock versus group	1.00	606	−3.56	−3.58	−1.62	−1.51	5.7	5.1

Table 2: Formation evidence and persistence, $H = 6$ months. $\hat{\pi}_0$ is computed over all spreads.

Out-of-sample, selected top-decile spreads decay rapidly toward null behavior. During trading windows, test statistics (τ_E) collapse from formation levels to between -1.6 and -1.75, approaching the simulated null baseline of -1.5. The re-test pass rate is 5.0% to 5.7% for pairs and stock-versus-group spreads, against about 5% under the null, and 5.9% to 9.0% for baskets. These findings confirm that stock spreads are at most weakly cointegrated, consistent with evidence that long-term cointegration persistence operates at or near test size [9, 10].

4.2 Net Returns

Table 3 confirms that no method yields profitable trading strategies. Top-decile portfolios incur annualized net losses of 0.28% to 0.97% in the training sample and 0.70% to 1.34% in the testing sample, with every testing-sample return proving statistically significant. These results align with previously documented losses in statistical arbitrage [6, 7, 8].

While null spreads lose approximately their transaction costs, selected spreads generally incur smaller losses. Pairs and stock-versus-group strategies yield near-zero lift over the null. In contrast, basket strategies generate a statistically significant positive lift of 0.4% to 0.9% annually, consistent with their slightly higher re-test pass rates. Nevertheless, because the null assumes independent Brownian motions, this lift may reflect unhedged equity risk factors or volatility clustering rather than true cointegration. Furthermore, formation p-values fail to reliably sort performance, as the top-minus-bottom return is not significant for any method in the testing sample.

Because the trading rule profits only from spreads that keep reverting, and formation raises the share of such spreads only from about 5% to 6%, cointegration-based formation cannot make

	Net return		Null	Lift		Top minus bottom	
Method	Mean	t	Mean	Mean	t	Mean	t
Training, 2010 to 2019							
Pairs	−0.97	−2.27	−1.12	0.15	0.35	0.27	0.43
Johansen	−0.28	−1.17	−1.07	0.79	3.13	1.67	3.56
PCA	−0.62	−3.58	−1.42	0.81	4.58	−0.40	−0.81
Eigenanalysis	−0.76	−3.24	−1.57	0.81	3.49	0.43	0.65
Stock versus group	−0.91	−3.20	−1.12	0.21	0.71	1.70	2.68
Testing, 2020 to 2025							
Pairs	−1.34	−3.07	−1.12	−0.23	−0.52	−0.14	−0.14
Johansen	−0.70	−3.89	−1.11	0.41	2.46	−0.07	−0.18
PCA	−0.84	−3.62	−1.46	0.61	2.39	0.20	0.38
Eigenanalysis	−0.78	−3.60	−1.70	0.91	3.51	0.59	0.86
Stock versus group	−1.06	−3.93	−1.14	0.07	0.27	2.07	1.48

Table 3: Annualized net return (%) of the top-decile portfolio (4), $H = 6$ months. Lift is the net return minus the null, and top minus bottom is the difference from the bottom-decile portfolio.

the spreads profitable. This agrees with Do and Faff [23], who attribute about 70% of the decline in pairs trading profits to non-convergence. As illustrated in Figure 1, these losses accumulate steadily rather than stemming from localized tail events, with the first half of 2020, which includes the market crash, losing 0.41% on average across the 5 methods.

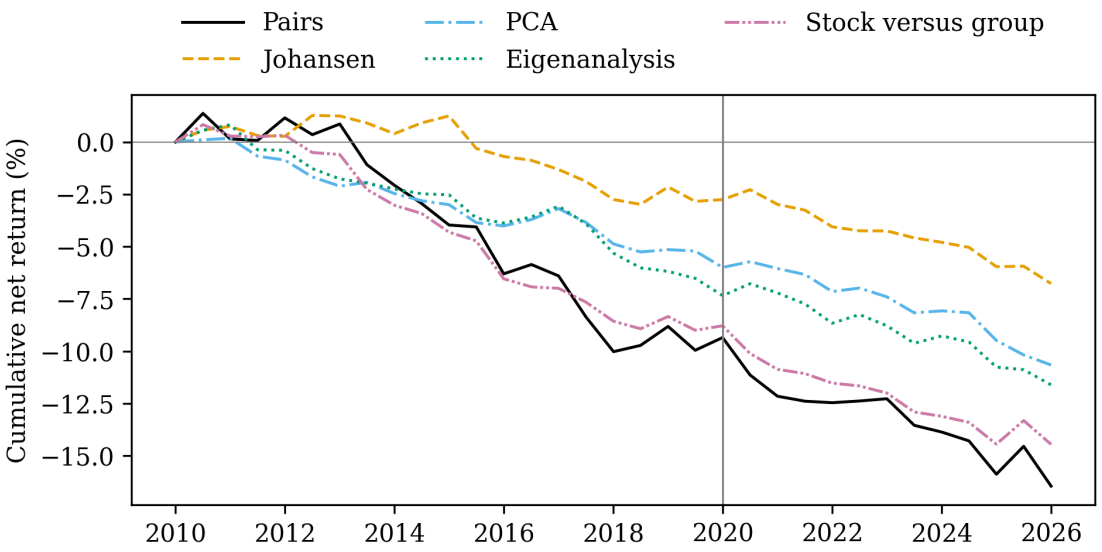


Figure 1: Cumulative net return of the top-decile portfolio (4) by method, $H = 6$ months. The vertical line marks the start of the testing sample.

4.3 Windows and Grouping

Table 4 confirms that performance remains uniformly negative across parameter configurations. Across all 40 combinations of formation windows (L) and trading horizons (H), average net returns across methods are consistently negative and mostly statistically significant. One-month trading

horizons incur the steepest losses, and relative performance across formation windows exhibits no stable pattern across horizons or sample periods.

	Training, 2010 to 2019				Testing, 2020 to 2025			
L (years)	$H = 1$	$H = 3$	$H = 6$	$H = 12$	$H = 1$	$H = 3$	$H = 6$	$H = 12$
1	−1.79	−1.14	−1.12	−1.22	−1.49	−1.11	−0.48	−0.89
2	−1.15	−0.59	−0.54	−1.46	−1.49	−1.29	−0.92	−1.33
3	−1.08	−0.90	−0.92	−0.98	−1.79	−1.39	−1.57	−1.14
4	−1.03	−0.75	−0.55	−0.72	−1.07	−0.90	−1.04	−0.38
5	−1.21	−1.04	−0.67	−0.46	−1.12	−0.85	−0.82	−0.31

Table 4: Annualized net return (%) of the equal-weighted average of the 5 method portfolios over the three groupings, by formation window L and trading horizon H in months.

Grouping choice similarly fails to achieve profitability, though cluster-based spreads lose less than industry sectors. With $H = 6$ months, sectors lose 0.86% a year in the training sample and 1.33% in the testing sample, compared to 0.50% and 0.60% for OPTICS clusters, and 0.73% and 0.98% for complete-linkage clusters. OPTICS clusters significantly outperform sectors by 0.73% per year out-of-sample, whereas complete-linkage clusters yield a smaller, statistically insignificant advantage. These findings align with recent evidence that clustering algorithms offer at best marginal enhancements over standard sector classifications [24, 25].

5 Conclusion

This study investigates whether cointegration-based selection identifies profitable US large-cap spreads out-of-sample across 5 combination methods, sector and cluster groupings, 1- to 5-year formation windows, and 1- to 12-month trading horizons, benchmarked against independent random walk simulations.

The empirical evidence indicates that cointegration formation fails to select profitable strategies. In the 2020 to 2025 testing window, top-decile portfolios for every method generate annualized net losses between 0.70% and 1.34% after accounting for trading and borrowing costs. Formation p-values fail to consistently rank returns, and adjustments to windows or groupings do not alter the negative outcome. Moreover, out-of-sample mean reversion persists only marginally beyond random walk expectations, aligning with evidence attributing pairs trading degradation to non-convergence [23].

While basket strategies exhibit positive lift relative to the random walk null, this likely reflects uncaptured common factors or volatility clustering rather than true cointegration. Robustness checks in Appendix B confirm that these negative returns persist across alternative exit thresholds and cost assumptions.

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A Methodology Details

A.1 ADF Statistic

For a series s_t, the ADF regression with a constant and one lagged difference is

$$\Delta s_t = a + \gamma\, s_{t-1} + \phi\, \Delta s_{t-1} + \varepsilon_t, \tag{5}$$

and $\tau = \hat{\gamma}/\operatorname{se}(\hat{\gamma})$. A more negative τ is stronger evidence that s_t is stationary.

A.2 Grouping

Market mode. With I_t the equal-weighted mean of the daily log returns of all eligible stocks, each stock's return is regressed over Φ as in [13],

$$\Delta \ln P_{i,t} = \mu_i + \kappa_i I_t + \eta_{i,t}. \tag{6}$$

Both cluster groupings use the residuals $\eta_{i,t}$ of the m stocks in a sector.

OPTICS clusters. Let C be the Pearson correlation matrix of the residuals. Eigenvalues of C above the upper edge of a random correlation spectrum [14],

$$\lambda_+ = \left(1 + \sqrt{m/T}\right)^2, \tag{7}$$

give J eigenvectors $e_1, \ldots, e_J$, and stock i is represented by its loadings $(e_{1,i}, \ldots, e_{J,i})$. OPTICS clusters these loadings with a minimum of 5 points. A sector with $J = 0$, or with fewer than 5 stocks, forms one group.

Complete-linkage clusters. Let ρ be the Spearman correlation matrix of the residuals. The tree of complete linkage on the distance δ_{ij} is cut into K clusters, where

$$\delta_{ij} = \sqrt{2(1-\rho_{ij})}, \qquad K = \max\big\{1,\ \#\{\text{eigenvalues of } \rho > \lambda_+\}\big\}. \tag{8}$$

A.3 Combination Weights

Let $\tilde{p}_t = (\tilde{p}_{1,t}, \ldots, \tilde{p}_{m,t})^\top$ be the normalized prices of a group over Φ with mean $\bar{p}$. Each method gives a vector v, normalized as

$$w = \pm \frac{v}{\|v\|_1}, \tag{9}$$

with the sign chosen so that the largest weight in absolute value is positive.

Pairs. For stocks i and j,

$$\hat{\beta}_{ij} = \frac{\widehat{\mathrm{Cov}}(\tilde{p}_i, \tilde{p}_j)}{\widehat{\mathrm{Var}}(\tilde{p}_j)}, \qquad v = (1, -\hat{\beta}_{ij}), \tag{10}$$

and likewise with i and j exchanged. The ordering with the smaller τ is kept.

Johansen. The vector error correction model

$$\Delta\tilde{p}_t = \nu + \Pi\tilde{p}_{t-1} + \Gamma\Delta\tilde{p}_{t-1} + \varepsilon_t \tag{11}$$

is estimated by reduced rank regression [3]. With $R_{0,t}$ and $R_{1,t}$ the residuals of $\Delta\tilde{p}_t$ and $\tilde{p}_{t-1}$ on $(1, \Delta\tilde{p}_{t-1})$, and $S_{ab} = T^{-1}\sum_t R_{a,t}R_{b,t}^\top$ for $a, b \in \{0,1\}$, the eigenvalues $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_m$ solve

$$\left|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}\right| = 0. \tag{12}$$

The vector v is the eigenvector of $\hat{\lambda}_1$, and the formation statistic is the trace statistic $\Lambda = -T\sum_{i=1}^{m}\ln(1-\hat{\lambda}_i)$.

PCA. The vector v is the eigenvector of the smallest eigenvalue of the covariance of price levels [4],

$$\hat{\Sigma}_0 = \frac{1}{T}\sum_t (\tilde{p}_t - \bar{p})(\tilde{p}_t - \bar{p})^\top. \tag{13}$$

Eigenanalysis. The vector v is the eigenvector of the smallest eigenvalue of

$$\Omega = \sum_{h=0}^{5} \hat{\Sigma}_h\hat{\Sigma}_h^\top, \qquad \hat{\Sigma}_h = \frac{1}{T}\sum_t (\tilde{p}_{t+h} - \bar{p})(\tilde{p}_t - \bar{p})^\top, \tag{14}$$

with 5 lags as in the simulations of [16].

Stock versus group. For stock i, with $\bar{p}_{-i,t}$ the equal-weighted mean of the other $m-1$ stocks,

$$\hat{\beta}_i = \frac{\widehat{\mathrm{Cov}}(\tilde{p}_i, \bar{p}_{-i})}{\widehat{\mathrm{Var}}(\bar{p}_{-i})}, \qquad v_i = 1, \quad v_j = -\frac{\hat{\beta}_i}{m-1} \text{ for } j \neq i. \tag{15}$$

B Exit Levels and Costs

Table 5 repeats the equal-weighted average of the 5 method portfolios, $H = 6$ months, with the exit condition $z_t\, n_{t-1} \geq 0$ of (3) replaced by $z_t\, n_{t-1} \geq -0.5$ and $z_t\, n_{t-1} \geq 0.5$, and with costs of 5 and 25 bps. Relative to the exit at the mean, the exit level changes the return by at most 0.17% a year, and every variant loses money in both samples.

		Training		Testing	
Exit at $\lvert z_t \rvert$	Cost (bps)	Mean	t	Mean	t
0.5 before the mean	5	−0.38	−2.02	−0.56	−3.76
	10	−0.76	−3.93	−0.96	−6.45
	25	−1.91	−8.80	−2.18	−12.64
0 at the mean	5	−0.36	−1.89	−0.58	−4.44
	10	−0.71	−3.62	−0.95	−7.04
	25	−1.75	−8.20	−2.04	−12.61
0.5 past the mean	5	−0.37	−1.89	−0.55	−4.53
	10	−0.69	−3.47	−0.89	−7.09
	25	−1.66	−7.77	−1.91	−12.58

Table 5: Annualized net return (%) of the equal-weighted average of the 5 method portfolios under alternative exit levels and costs, $H = 6$ months.